Final Project: Pick a Project

Computer Science - Week 10

Oct 3, 2026 - Version 0.0.2

Please make sure to your UD **email** AND **name** is below. Unless your UD emails are included in this table, then you will not earn any points for this assignment when it is graded! **Every person should submit their own individual worksheet**.

Talk about each of the questions with a friend or our LLM (or both), but **make sure the words you enter below are yours** (with the obvious exceptions notwithstanding). Brainstorming is best done in full sentences, and the main goal here is brainstorming. It's worth reviewing the information in this series of pages in the Drafter documentation to gain a better understanding of the process.

Role	Name	Email
Developer (You)	Derek Bellin	derekbel@udel.edu
Friend (optional)		

Part 1: The Goal & Prior Implementations

1. **Identify the Goal:** What specific goal are you trying to address? Think about your passions, hobbies, and academic interests. The latter are an excellent candidate to focus on. Some majors have suggestions listed below: these were provided by faculty in those departments. Other majors could consider a focus tied to a big final assignment in one of their recent courses in their major.
    - **Biomedical Engineering** majors may consider one of the following:
        - If you have started (or completed) the Cell/Tissue Design final project in BMEG100, design an organoid, which involves combining a cell type, biomaterial, and testing method. Have the website include a description and diagram of your design concept, a testing method, a needs statement, and an impact statement relevant to their organoid. Also include a link to a paper that supports your idea and a paragraph describing how that paper supports your design concept.

        - Otherwise, create a website related to your BMEG100 Lab Report 1, *Cytoskeletal and Nuclear Staining Lab Report Guidelines*, linked here (University of Delaware account required to access).
    - **Chemistry** and **Chemical Engineering** majors may consider creating an automated chemistry assistant that takes balanced reaction equations, starting reactant masses, and a harvested product amount to determine the limiting reactant, theoretical yield, and percent yield based on principles from Chemistry 2e (OpenStax) — Chapter 4: Chemical Reactions and Stoichiometry. To avoid manually compiling atomic mass tables or writing custom string tokenizers, use the ChemPy library to parse molecular formulas into exact molecular weights and verify reaction stoichiometry against compound data from the NIH PubChem Database. The program outputs an organized breakdown of limiting reagent status, product yields, and remaining masses of unreacted species. Also use the heat of the reaction to determine if it is exothermic or endothermic, and compute the heat flow per mole of the product.
    - Majors in the **Electric & Computer Engineering** department may consider a steganography project that hides messages in images, and then retrieves those messages from uploaded images.
    - **Mechanical Engineering** majors may consider a project that somehow uses both the NumPy and SciPy Python libraries.

My goal is to create a Basketball Shot Predictor website that uses physics to predict if a shot will go in. The user will input the shot angle, the starting height, the force/ ball speed, and the distance from the hoop. Using NumPy and SciPy, the program will calculate the ball's path and show a graph of its trajectory. The user will be able to change the data and recheck if the shot will go in. This project combines my love of sports and my interest in physics, while providing a challenge in programming.

2. **Prior Implementations:** Search the internet. What existing tools or websites currently attempt to address it?

Some similar programs I found are…

1. *SkyMotion Calculator*
    a. *It uses launch speed and angle to display a graph of the ball's path*
2. *Basketball Shot Arc Sim*
    a. *Lets the user change distance, shot angle, and height to see how the ball will travel*
3. *Projectile Motion Animator*
    a. *Allows the user to enter speed, angle, and height to view trajectory*

3. **Your Implementation:** Briefly describe the legitimate purpose or functionality of your site. How would you want this web application to be used?

I want my website to be a useful feature to help predict what angles, speed, and curve are most effective when taking a basketball shot. I also want it to be used to see if a trick shot is possible with limited strength and force. Also, later I may want to add a defender or object to help the user see if the shot would get blocked or what a deflection would look like.

Part 2: The Input-Process-Output (IPO) Model

Think about the fundamental flow of your program.

1. **The Core Goal:** Every program has three main phases: taking input, processing data, and presenting output. In one sentence, what is the ultimate goal your program will accomplish for the user? Be specific and use CS1 concepts.

The program will use user inputs and physics calculations to predict whether a basketball shot will go in.

2. Input: What specific information will the user need to provide to your program? (e.g., a city name, a numerical budget, a list of hobbies).

The user will enter the ball's speed, launch angle, starting height, and distance from the hoop.

3. Processing: At a high level, how will the computer manipulate or evaluate that data? (e.g., calculate the average, filter out options that are too expensive, compare the input against a database).

The program will use NumPy and SciPy to calculate the ball's trajectory and compare it to the hoop's location and the defender's height.

4. Output: What will the user get back? (e.g., a recommendation, a calculated safety factor, a formatted schedule).

The user will see a graph of the ball's path and a prediction of whether the shot will go in, go above the hoop, or fall below the hoop.

Part 3: Data Abstraction

Identify the "nouns" of your problem.

1. **Representing Reality:** Abstraction is the process of representing the real world inside a computer by ignoring unnecessary details. What are the core "things" (nouns) your program needs to track? (e.g., *If building a recipe app, the nouns are "Recipe" and "Ingredient"*).

The main things my program needs to track are the basketball, the hoop, and the ball's trajectory

2. **Identifying Properties:** For the most important "thing" you listed above, what 3 to 4 specific details do you actually need to solve your problem, and what details will you abstract away?
    - *Details I need:* (e.g., *Ingredient name, weight, cost*)
    - *Details I will ignore:* (e.g., *Ingredient color, brand, expiration date*)

Details: Ball speed, launch angle, starting height, and distance from the hoop *Details I will ignore: ball color, player appearance, brand of basketball, and court design*

Part 4: Challenges, AI Strategy, & Goals

1. **Difficulties:** What parts of this project seem the most difficult or technically complex?

The hardest part will be accurately calculating the ball's trajectory and determining whether it will go in the hoop once it gets there.

2. **Coding Strategy:** If you will be using an LLM to help code, how will you ensure you understand the generated code and avoid "hallucinations" or "shallow learning" (copying without understanding)? If you won't be using an LLM to help code, this project might take about 4 hours a week to complete, so list each of the days and times in which you work on the project.

I will use AI to help with coding, but I will look over the code to understand how it works. I will ask for explanations and ask for tests to understand how everything works.

https://derekb-udel.github.io/Accuracy-Shot-Site/–about

3. **Goals:** What is your ultimate goal for this project? (e.g., *Impress the graders with a creative UI, build something I can use in my other classes, learn Drafter deeply*).

My goal is to have fun with this project while making it interactive and useful to younger kids who have fun seeing if their 'trick' shot is possible. I also want to learn more about drafter and how to use the website to the fullest.

Part 5: Research & Brainstorming Conversation

Have a conversation with a Generative AI tool (like ChatGPT or Claude) OR a friend about your project idea. Do not ask for code. Instead, use the AI/friend as a sounding board to research your problem.

1. **The Initial Prompt/Question:** Ask the AI or your friend to help you brainstorm. (e.g., *"I want to build a simple web tool to help students track their study hours. What are 3 common problems students face when tracking their time, and what are 3 simple features a basic tool should have?"*). Paste your prompt below:

I want to create a Basketball shot predictor using physics. What are some useful features I could include, and what spins can I put on my code to make it more interesting and useful?

2. **The Response:** Briefly summarize the best ideas the AI or your friend suggested. What existing solutions or prior apps did they mention?

The AI suggested letting users enter the ball's speed, angle, starting height, and distance from the hoop. It also suggested showing a graph of the ball's trajectory. Finally, it suggested adding obstacles that change the shot from a linear to a more indirect route.

3. **The Pushback:** You must reply to the AI/friend to narrow the scope when it's over-ambitious. Pick the most complex feature they suggested and explain why it is too complicated for an introductory programming project. Ask them for a simpler alternative. *Paste your response here:*

Predicting a real basketball shot might be too complicated for this project. Could I simplify it by ignoring air resistance and focusing on the speed, angle, and distance?

4. **Reflection:** Based on this conversation, how has your project idea changed or become more focused?

My project is now more focused on using basic physics to predict basketball shots. It will focus on the principle of whether the shot is possible and ignore external factors that will affect the shot's performance momentarily.

Part 6: Resource Planning

1. What resources will you use if you get stuck?

I will look at projects similar to mine and see how they overcame the challenges. I will also use AI and ask for its assistance for what my error or problem might be.

2. If your LLM provides code that causes a Drafter or Bakery error, or if you run into a problem during deployment, what is your step-by-step debugging plan to resolve it?

I can run it through Thonny and use the line-by-line tool ot see at what point the project breaks. I can also use AI to describe each step to me and explain what it is doing.

Part 7: LLM Conversation

If you used an LLM in any part of this work, copy and paste every part of that conversation below. Otherwise, you may be given a zero for the assignment. Obviously, these words will not all be yours, and this is the exception to the requirement at the top of this assignment that all responses be in your own words. If you did not use an LLM to assist you in this assignment, simply say so below.

I used ChatGPT to help brainstorm my Basketball Shot Predictor project and identify similar programs. I also used it to identify possible challenges, simplify the project scope, and explain how physics and Python could be used to predict a basketball shot.

Part 8: Submit

In Google docs, with this document open, please click on *File* > *Download* > *Microsoft Word (.docx)*, and make sure you put that download somewhere you can find it again. Then, upload it to the Canvas assignment about selecting or brainstorming your final project.

When you're done with this assignment, review this page of the Drafter documentation to better prepare for the next step.